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Workforce forecasting models: A systematic review

Anahita Safarishahrbijari

Department of Computer Science, University of Saskatchewan, Saskatoon, Saskatchewan, Canada

Correspondence
Anahita Safarishahrbijari, Department of Computer Science, 110 Science Place, University of Saskatchewan, Saskatoon, SK S7N 5C9, Canada.
Email: anahita.safari@usask.ca

Abstract

Workforce analytics involves using models that integrate internal and external data to predict future workforce and help organizations in any industry examine factors that have a prognostic effect. This paper assesses workforce modeling and prediction methods by examining their rationale, strengths, and constraints. It aims to identify enhancements for further development of workforce forecasting models and compares the capacity and reliability of different forecasting methods. Past and present modeling trends are described and critiqued based on their relevance to current requirements. Several approaches are reviewed, such as time series modeling and system dynamics simulation. Sensitivity analysis in models is assessed. The models are decomposed into three modes: supply-based, demand-based, and need-based, which in some cases provide substantially different estimates of future workforce need. The chronological progression of models' development is analyzed. The articles are also classified based on the countries and the sectors that have paid great attention to workforce prediction research. Consideration of the use of workforce models and the inputs into such models is not within the scope of this paper.

KEYWORDS

labor forecasting, manpower planning, organizational studies, workforce, workforce modeling

1 | INTRODUCTION

Workforce supply and demand modeling can help inform human resource (HR) planning and decision making. One of the principal reasons to use workforce forecasting models is to identify the potential gap between the present workforce and future requirements in order to predict training and recruitment needs. The question as to whether the statement "Rising labor costs in labor-intensive industries" is true has increased the concern about the cost effectiveness of the delivery of services. Since workforce models can involve a high degree of complexity, tradeoffs across multiple objectives, and numerous uncertainties, the forecasting and modeling methods in this area have evolved over time, becoming more complex and taking into account external forces and factors. Depending on the purpose of the model and the questions to be answered, system factors, data requirements, and modeling methods will vary. The scope of such models can be categorized into three major perspectives: supply based, demand based, and need based. In supply-based models the goal of projection is determining the number of staff available at a particular time or period of time based on training, retirement, and promotion or demotion. Supply-based models provide a response to "How many providers will there be?" Demand-based models deal with prediction of future service requirements and likely changes in the demand and provide an answer to "How many supplies will we need?" In addition, needs-based estimates use an exogenous benchmark to judge the adequacy of the number of staff required to meet the defined targets. Need-based models respond to

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"How many suppliers should we need?" Models may be composed of one or more of these components, depending on the objectives of modeling. Classifying workforce planning models captured in this article into these three major categories, we summarize published models and review their capabilities and potential limitations.

This article presents a systematic literature review of the operational research approaches applied to workforce projections and categorizing them into several major branches based on modeling methodologies: qualitative, time series, regression models, simulation, optimization, and generic mathematical models. The intention of this review is to investigate methodological strengths and challenges of previous workforce models and discuss methods for improving future models.

2 | PREVIOUS REVIEWS

Some authors have synthetized modeling techniques for workforce projections in a general way before our work. Edwards (1983) reviewed models developed in the UK, focusing their underlying assumptions and applications rather than concentrating on statistical and mathematical details. Lomas, Stoddart, and Barer (1985) critically reviewed the physician forecasting approaches in Canada and outlined the deficiencies inherent in Canadian techniques in comparison with an extensive research by the Graduate Medical Education National Advisory Committee (GMENAC), an advisory group to the US Department of Health and Human Services. Smith and Bartholomew (1988) explored the chronological growth of workforce predictive models in the UK from World War II to 1985 and reported organizations such as Civil Service's Centre for Administrative Studies, which laid the basic structure for workforce planning in that country. Prescott (1991) classified the approaches used in the projection of healthcare workforce requirements into five groups: (I) the professional needs approach, which is based on professionals' opinion; (II) personnel-to-population ratio, which considers changes in population and service delivery; (III) econometric models based on the ability of people to purchase services; (IV) the Bureau of Labor Statistics technique, within which the demand is estimated as the end result of several stages based on separate models; and (V) hybrid models, which are a combination of previous methodologies. Warner and Asch (2000) reviewed existing contributions to military workforce planning. They discussed challenges arising from rapid alterations in the duties and roles of military workforce and the civilian labor market. O'Brien-Pallas et al. (2001) reported the forecasting methodologies for nursing requirements from 1996 to 1999. They elaborated the shortcomings associated with existing models such as ignoring complex fundamentals that affect human resources for health (HRH), including the effect of price competition on workforce participation and ignoring the influence of HRH decisions on public health. Furthermore, they discussed need-, demand-, and utilization-based approaches and mentioned that the estimates for future nurse requirements based on each of these three approaches vary considerably. Roberfroid, Leonard, and Stordeur (2009) carried out a literature review on physician supply models and classified forecasting models into four genres: supply projections, demand-based approaches, need-based methods, and benchmarking among similar health systems. They identified the vast majority of models as static models that lacked historic data to test their validity. In a comprehensive review by Wong, Chan, and Chiang (2012), recent workforce planning methods were assessed to identify improvements for future workforce models developed in the construction industry. They distinguished top-down forecasting as the chief approach used for dealing with workforce demand rather than bottom-up methodologies. Top-down, also known as stepwise models, are formulated based on a broad view of the whole system without describing the subsystems in detail. This approach assumes that growth in a particular industry results in proportionate growth for each position in that industry. In contrast, a bottom-up approach merges first-level subsystems to bring about more complex systems.

Our review expands on these previous reviews as it includes workforce forecasting models in all industries, including an analysis of methodologies and the quality and quantity of publications in the area of workforce projections.

2.1 | Methods

2.1.1 | Paper selection and data derivation

We systematically reviewed the literature based on the following keywords: Workforce, Manpower, Personnel, Human resource, Predict, Plan, Forecast, Project, Simulate, and Model. We searched the following databases, restricting the results to journal articles published between January 1980 and May 2015: Elsevier (SCOPUS and Science Direct), Springer, Emerald, Wiley, Gale, and MEDLINE/PubMed. We omitted papers based on organization design, training and hiring procedures, personnel and outsourcing cost evaluations, reassigning personnel, the flexibility of personnel, transition strategies, labor turnover, and human resource allocation based on titles and abstracts. When necessary, a full text review was conducted. After full text skimming,

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forecasting models for work schedules, multipurpose models, and non-English models were left out. Some conference papers, particularly from the Winter Simulation Conference, a dissertation, and some studies from the reference list of articles, were included, and in the end 275 papers were included in our results. The paper selecting procedure is demonstrated in Figure 1.

2.1.2 | Paper classification

Papers were classified by country, industry, scope, year, and modeling techniques. They were categorized by country where a substantial number of papers were found for country specifications, otherwise they were grouped based on continent. Workforce planning studies can be explored through three scopes considering forthcoming need to workforce, future demand for workforce, and prospective supply of human resources. *Need* is the volume of services required to provide a desired standard of services (Turner, Ostby, & Linda, 1993). Need-based models provide a target level for services independent of those actually being provided through the marketplace (Jacoby, 1995). *Demand* is described as the volume of services being requested but not necessarily required. Demand is subject to cultural and ethical causal factors of current society as well as social and educational influences such as media, which affect willingness and ability to seek services. It also depends on geographical access, quality, and price of services. *Supply* is defined as services that can be provided, and is vulnerable to political and public pressure and historical trends. Supply, among other factors, deals with current workforce resources, the current style of the labor market, upcoming HR entry and attrition, and future working conditions. In a perfect system, need for a service should equal demand of that service over time, but this is never the case. For example, in healthcare systems, individuals who are in real need to care service might never demand it for various reasons such as education and insurance policies. On the other hand, some individuals who demand health services might not really need them. Figure 2 is a scheme of the above explanations. Need, demand, and supply are represented in three different ovals and the intersections are specified by alphabetic letters: A, B, C, D, E, and F. One of the prominent aims of workforce predicting is to settle the differences between demand, need, and supply, by analyzing training, labor market entries and exits, remunerations, and so on—that is, increasing the overlap of these three scopes as shown by area 'C' in Figure 2.

To examine included articles, we considered two topics—

1 We seek to classify existing research to determine industries, countries, and methodologies used.
2 We analyze the quality of the models developed by investigating models testing, sensitivity analysis and model implementation.

—and find responses to our research questions:

- Which industries have been targeted for workforce modeling research?
- Which countries have paid attention to workforce projection issues?
- How have models' quality and quantity progressed over time?

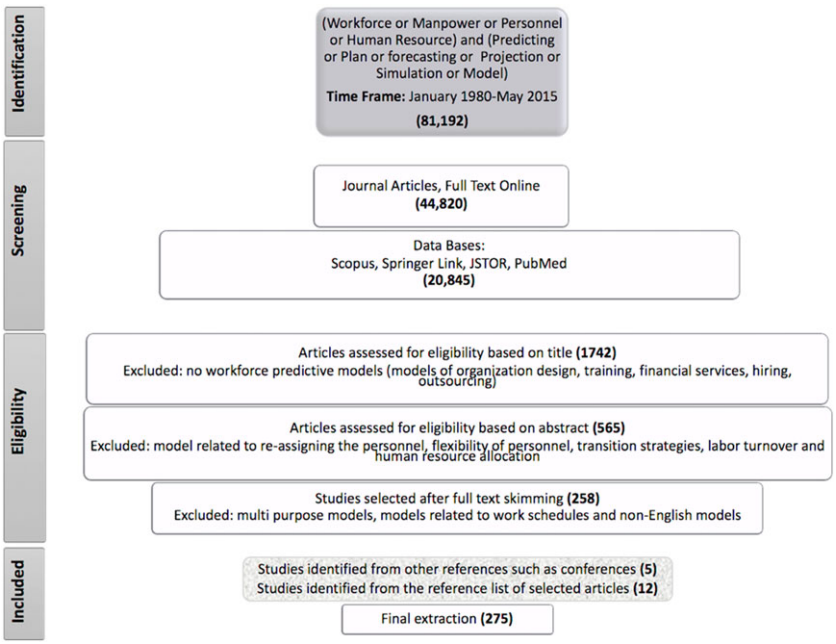

FIGURE 1 PRISMA diagram showing the flow of studies [Colour figure can be viewed at wileyonlinelibrary.com]

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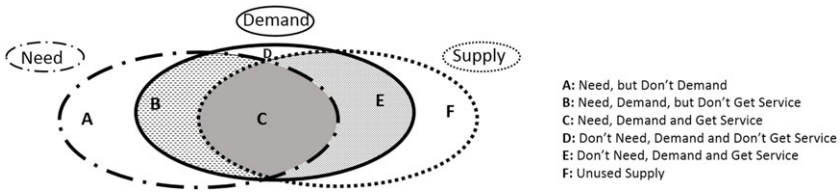

FIGURE 2 Intersection of need, demand, and supply [Colour figure can be viewed at wileyonlinelibrary.com]

- What are the dominant techniques for workforce modeling?
- Has the validity of models been tested or not?
- To what extent has the sensitivity analysis been considered in workforce systems modeling?
- Have the models been used in real world?

2.2 | Summary statistics

In this section, we provide some interpretations based on analysis of chart and frequency tables. We also investigate relations between methods and scopes.

Figure 3 shows the outcome of studies found relating to different industries with the industries represented categorized under eight headings: Healthcare, Construction, Industry, Service Industries, Military and Navy, Agriculture and Horticulture, Marine and Airlines, and General HR. Models related to library operations, call centers, postal service, information security, urban police, hotels, banking, criminal justice workforce, retail chains, air cargo terminals, civil services, and tourism industry are categorized under the heading "Service Industries." Models that do not belong to any specific industry have been classified under the heading "General HR." Slightly more than half of the articles fall into the Healthcare category. General HR, Industry, and Service Industries are the next most frequent areas for which workforce models have been developed. Due to Healthcare's dominance these articles were further classified by workforce type into Nurses, Physicians and Surgeons, Other Staff, and General Healthcare. The heading "Other Staff" includes staff working in hospital laboratories, emergency department providers, operating room teams and veteran affairs cadre. General Healthcare includes models that do not represent a specific group of healthcare workforce—for example, comprehensive models considering nurses, surgeons, and other staff working in a particular health department. As shown in Figure 3, the majority of healthcare models are focused on physician and surgeon workforces.

When examined based on the country studied (Figure 4), it becomes clear that the USA leads in workforce modeling, while Canada, the UK, and Australia and New Zealand hold second, third, and fourth most frequent publications, respectively.

The volumes of studies by year of publication (Figure 5) indicate that the prevalence of models in 5-year periods increases about fourfold between 1980 and 1984 and 2010–May 2015.

It should be noted that the total number of articles in a particular scope which has been reported in the first row of Table 1 is not equivalent to the sum of articles in the relevant column, because in some works hierarchical modeling was employed. For example, some authors have modeled demand using gray theory but employed Markov methods for the supply side (Lin, Wang, & Chiang, 2001), whereas others have combined Markov models and goal programming (Zanakis & Maret, 1981).

Models within each type were further classified by method (Table 1). "Others" refers to the infrequently used approaches such as queuing theory and Atkinson's model. Adaptive search algorithms such as genetic algorithms (GA) and neural networks were also used by a couple of authors for workforce predicting (Alba, Luque,

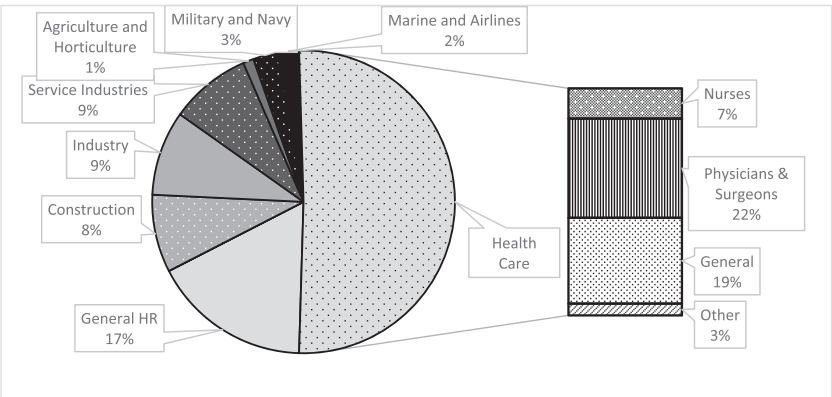

FIGURE 3 Percentage of articles related to different areas

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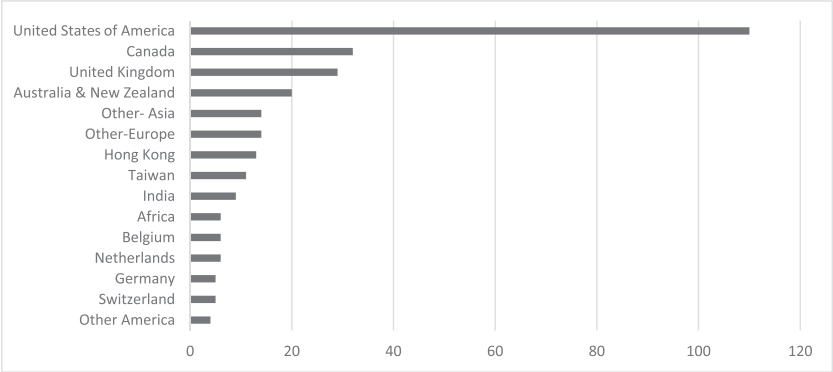

FIGURE 4 Volume of publications based on country studied

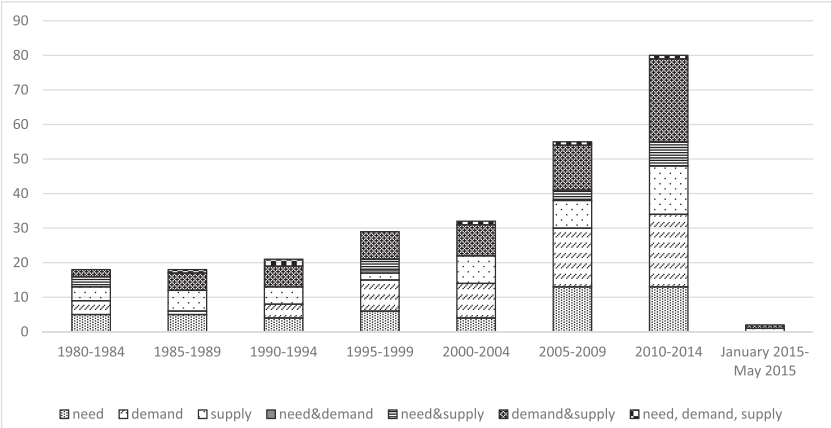

FIGURE 5 Number of articles published in 5-year periods (need, demand, supply)

TABLE 1 Volume of studies based on use of different modeling approaches and scopes

Modeling approach		Need based	Demand based	Supply based	Supply and demand	Supply and need	Demand and need	Supply, demand, and need
Qualitative		10	2	9	18	5	2	3
Time series	Markovian	2	1	3	3			1
	Other time series		9	2	1			
Optimization		7	11	7	3	1		
Statistical and regression		9	13	7	6	2		1
Simulation	System dynamics	3	7	4	7	2		1
	Other simulations	7	7	5	6	3		
Analytical stock and flow				3	3			
Generic mathematical modeling		4	10	8	9	2		
GA: neural networks, fuzzy and grey analysis		3	3		4			
Others		5	6	1	7	2		
Not identified		2	2	3	8			
Grand total		52	71	52	75	17	2	6

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& Luna, 2007; Bee-Hua, 2000), as were approaches based on partial truth concepts such as gray modeling and fuzzy logic (Choudhury, Sarkar, & Mukherjee, 2002a; Ho, 2010; Lin & Juan, 2010). These techniques have been used solely or hierarchically with other approaches in workforce forecasting models. As these approaches are not widely used, they are not discussed in further detail in this review.

It can be noted that the models simultaneously considering demand and supply are most common, while models considering just demand are the second most frequent models. Models that examine need and supply or all three components (need, demand, and supply) are the least widespread. It can also be observed that most of the methods are not being used in a specific mode but, for example, time series techniques are mostly used in demand-based models.

3 | REVIEW OF MAJOR MODELING APPROACHES

Focusing on the most frequently used modeling approaches, we first introduce the mathematical framework underlying the model construction, then overview the published applications in different industries, and finally discuss the potential limitations of the techniques involved.

3.1 | Qualitative models

Qualitative methods are the simplest approaches in workforce planning. These methods are particularly useful when high-quality empirical data are not available. Instead, they rely heavily on the experience of engaged experts. Qualitative methods used in HR modeling include exploratory or descriptive methods that scrutinize the behavior of an organization and represent deterministic relationships in variables of a survey (Shemin et al., 2002), interviews (Masnick, 2009), and census (Hagopian, Mohanty, Das, & House, 2012).

A discrete-choice experiment (DCE) is a quantitative technique for drawing out individual preferences and converting qualitative data into useful computational data. This method enables researchers to find how individuals judge selected attributes by asking them to state their choice over different hypothetical alternatives. Although DCE methods are very useful in situations that are purely hypothetical, they are complex research means that require expertise in a number of different areas (Lagarde & Blaauw, 2009).

The Delphi method is another popular consensus-based method in which a panel of experts typically answer questionnaires over several iterations. After each iteration, an anonymous summary of the results is processed and provided as the basis of the next iteration, in which they may revise their earlier judgments in light of the answers of the other experts. The goal is to eventually reach a unanimous, and hopefully correct answer (Bryant, Maggard & Taylor, 1973). The Delphi method has been used in a variety of workforce modeling efforts to determine the unknown parameters of the models or has been used hierarchically in combination with the other modeling techniques. The Graduate Medical Education National Advisory Committee (GMENAC), an advisory group to the US Department of Health and Human Services, used this method along with mathematical modeling to analyze the future physician need and supply by specialty and geography, and explored the impact alternative policies could have on balancing these two sides (McNutt, 1981; Morgan, 1982). Kwak, McCarthy, and Parker (1997) also used a Delphi process to identify all factors affecting a supply-and-demand model of clinical laboratory personnel in an urban academic health center and used these factors in an analytic hierarchy processing model. Solnet et al. (2014) used a three-round modified Delphi technique to identify a long-term tourism workforce. They then used the results to run possible scenarios. Their modification to the Delphi technique was that they reviewed a broad range of tourism workforce literature to identify the factors most likely to have an impact on the tourism workforce. The expert panelists were then asked to judge the importance of these factors on the future of the Asia–Pacific tourism workforce.

Some criticisms of this method include its dependency on experts' opinions, which may not yield accurate estimations in the end. Another weakness associated with qualitative modeling is that it can be a time-consuming process, requiring a substantial amount of costly executive time (Parker & Caine, 1996). Qualitative models are also typically beneficial for small organizations (Lawrence et al., 2006). Nevertheless, the Delphi method and its revised versions are still being used by some researchers.

3.2 | Time series models

The time series approach is another popular method for forecasting future values based on historical data. Among time series modeling methods, Box–Jenkins, exponential smoothing, vector error correction and Markov modeling have been widely used in workforce forecasting applications.

Box–Jenkins uses an autoregressive moving average to find the optimum fit of a model to historic time series data (Commandeur & Koopman, 2007), and was used by Wong, Albert, and Chiang (2005) to forecast key

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indicators in the construction labor market of Hong Kong. They tested the accuracy of the model using Theil's *U* statistic and the mean absolute percentage error, concluding that most of their proposed forecasting models have reasonably good predictive performance.

Exponential smoothing is a weighted moving average methodology in which past observations are assigned exponentially decreasing weights over time; weighted patterns can easily be altered to meet special needs and forecast according to past cyclic predictions. An exponential smoothing method has been used to forecast the personnel trend in the energy technology industry in Taiwan from 2009 to 2014 based on data from 1997 to 2008 (Hsu, Chen, & Hsien, 2012).

Error correction models are dynamical multiple time series models; their short-term dynamics are fed from the dispersion of their current state and their long-term dynamics. These models estimate the rate of a dependent variable's return to equilibrium after a change in an independent variable. Wong, Albert, and Chiang (2007) took the advantages of this approach to forecast construction workforce demand in Hong Kong. Based on dynamic econometric modeling techniques, they found that the interrelated economic variables such as material prices, labor productivity, construction output, real wages, and bank rate (inter-bank interest offered on loans) have a long-run equilibrium effect on workforce demand. The model was tested against various diagnostic statistical criteria for forecasting purposes. Later, Wong, Ng, Chan, and Chiang (2009) developed the relationship between workforce demand and the effects found in their earlier work by applying vector error correction modeling, regression analysis, and exponential smoothing methodology.

Other types of time series models have been used to forecast call center staff requirements (Shen & Huang, 2008), and physician and nurse supply in Serbia (Santric-Milicevic et al., 2013).

A widespread time series approach is Markov analysis, which models movements among states based on transition probabilities. Trivedi, Moscovice, Bass, and Brooks (1987) developed a semi-Markov model to predict the supply of primary care providers in a given geographical region in the USA. Zeffane and Mayo (1994) discuss more complex models based on Markov, and Reid and Taylor (1989) applied an absorbing Markov analysis to model an educational administration workforce. They divided the staff into eight discrete states: newly employed, employed, retired, deceased, resigned, on sabbatical, on leave without pay, and ill for more than 30 days. Kinstler, Johnson, Richter, and Kocher (2008) used Markov analysis to address the supply of navy nurse corps. They investigated staff flow of junior nurses to Ensign, Lieutenant Junior Grade and Lieutenant within the navy nurse corp. In their model, the workforce system was managed through recruitment processes and hiring sources; promotion rates at the lowest two ranks were mandated, but not at the higher ranks, and retention rates varied across pay grades. Belhaj and Tkiouat (2013) introduced a supply-and-demand forecasting model of the number of employees in a hierarchical time-dependent human resource system. They assumed that each employee has limited promotion possibilities. Therefore, for a better result, they put individual employees into their own skill mix cohort where they could evolve. The authors' model reveals the gaps between forecasted results and management expectations.

The key weakness of Markov models is that they are not capable of interacting with their environment; for example, the entry and exit of entities in the system cannot easily be modeled. Also, Markov models are not capable of incorporating feedbacks in the system, such as the impact of promotion or wage on attrition (Bartholomew, Forbes, & McClean, 1991). Due to the mathematical underpinnings of the method, models of complex systems can only be analyzed via unique computer programs that are not easy for non-mathematicians to deal with. Thus such methods are not practical or easily understood by managers, as mathematicians usually do not tend to work as managers.

Although the underlying trends, cycles, periodic elements, and patterns of past data are considered in time series models, these approaches are disadvantaged by the assumption that recent trends will continue and the fact they ignore factors other than time. The other weakness of this approach is that there is always a need for historic data. Another limitation of these models is that their reliability depends on predictions of system behavior over time; hence they may be unsuitable or sometimes impossible for long-term runs.

3.3 | Optimization models

Optimization, which is concerned with minimizing and maximizing an objective function, usually subject to various constraints, has been widely used in workforce modeling. The most popular optimization approaches found among the included papers are linear programming (LP), goal programming (GP), mixed integer linear programming (MILP), and dynamic programming (DP). LP, also known as linear optimization, aims to find the optimum value of a linear function, subjected to linear equality or inequality constraints, and is a widely used technique in workforce optimization problems, whereas GP introduces multi-objective optimizations. MILP presents a linear objective function involved with a mix of linear and discrete integer constraints, whereas

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dynamic programming is a technique for solving multistage optimization problems in which a large problem is decomposed into smaller hierarchical problems.

Eiger, Jacobs, Chung, and Selsor (1988) developed a mixed LP–simulation model for the US military encompassing training, hiring, promotion, and retirement. They also used a personnel flow simulation model as the controlling input to stabilize the optimizer's results and for features of the system, which could not be modeled via LP. Srour, Haas, and Morton (2006) generated an LP scheme for efficient fit of demand and supply of construction personnel considering training, hiring, and allocation. Niehaus (1995) employed GP to model need and supply of a US navy shipyard community facing a wide-scale downsizing of personnel. Weigel and Wilcox (1993) combined GP, LP, and Markov inventory techniques to model the supply side of the US Army's enlisted workforce decision support system. MILP has been used in a demand-based model to optimize the size and arrangement of long-term employees in a postal service office (Qi & Bard, 2006), and to minimize the costs associated with the workforce size in a mail-processing center (Judice, Martins, & Nunes, 2005). Atlason and Epelman (2004) transformed a nonlinear personnel-planning system into an MILP to obtain the optimum level of staffing in a call center. Ganguly, Lawrence, and Prather (2014) used an MILP model to investigate the optimal number of emergency department staff to minimize total staffing cost for a given target service level for arriving patients. Wild and Schneeweib (1993) used a stochastic dynamic programming to optimize the simultaneous use of floaters, overtime, and temporary workers in a German service company as the case study. More recently, Özdemir-Akyıldırım and Talay-Değirmenci (2015) introduced a stochastic model and DP optimization tool for determining the optimal staffing of trainees under uncertainties such as attrition rates and study durations. They sought to minimize the deviations from the target class size in the last stage of the training program by adjusting the number of students admitted into the first stage. Based on numerical analysis, the resulting model was robust to the uncertain parameters.

Most of these studies sought the optimal solutions based on linear models of the systems and hence diverged from the nonlinear dynamic systems that exist in reality (Forrester, 1994). Yet numerical solutions for nonlinear optimization can be computationally complex. Thus they are not easy for nonmathematicians to analyze.

3.4 | Generic mathematical models

Mathematical models have been widely used to describe different features of the real world, their dynamics, and relationships through mathematics. Rapid progress of computational algorithms has led to development of even more sophisticated mathematical equations, since these algorithms help in treating mathematical models without the need to find explicit solutions to them. In one of the earliest works among the selected articles, Doran and Deen (1981) employed linear differential equations in workforce planning exploring the educated workforce requirements in the USA. However, after they examined both the sensitivity and stability of the model, they concluded that linear differential equations failed to describe the workforce system. Yet Jiang and Begun (2002) employed another linear differential equation to distinguish factors influencing physician supply in the USA. They took into account environmental variables such as hospital concentration and employer interests, but did not consider the effect of nonphysician providers such as physician assistants and nurses in their analysis. Recently, Sing, Love, and Tam (2012a) introduced a mathematical model to forecast labor demand in the construction industry. Based on a labor multiplier approach, they assumed a relationship between construction output and workforce demand. They also used a distributed lag model to help distinguish the relationship between a dependent variable (annual change in construction output) at a given point in time and an explanatory variable (annual change of gross domestic product) in times less than or equal to that given time.

Even though purely mathematical models are capable of capturing relationships between inputs and outputs of a system deterministically, they are impractical for characterizing historical or survey data. To better describe empirical data, researchers have mixed stochastic elements with deterministic ones and have developed stochastic and, more specifically, statistical models.

3.5 | Statistics and regression

A statistical model is a stochastic model that considers uncertain parameters in its projection of future incidents. The uncertain parameters are approximated based on historic data through a fitting process. The fitting process can be performed through a variety of statistical principles, such as least squares or maximum likelihood estimation. The characteristics of a statistical model and the measurements obtained from it are explored through variance, expected values, and other statistics. The best estimating principles are selected based on properties of the estimators such as interpretive ease, computational ease, mean squared error, bias, consistency, and variance (SAS Institute Inc, 2010).

As one type of statistical model, regression analysis investigates the causal effect of independent variables

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on dependent variables and is a powerful tool for exploring the interdependence of variables.

Among predictive methodologies, statistics and regression have been frequently used for a variety of forecasting purposes. Within the healthcare sector, they have been used to determine the optimum capacity of operating room on-call staffing for urgent procedures (Dexter & Liam, 2001), and model physician workforce requirements, taking into consideration elements such as physician work duties, patient service levels, and patient occupancy levels (Ben-Gal, Wangenheim, & Shtub, 2010). Lin and Juan (2010) developed a model focusing on industrial, organizational, and macroeconomic parameters affecting hotel human resource demand based on the generalized least square method. Regression analysis has also been used to approximate the workforce demand for real-state construction projects in India (Agarwal, Rajput, & Mangesh, 2013) and to examine the disaster mental health workforce requirements in Australia (Reifels, Naccarella, Blashki, & Pirkis, 2014).

Statistical models are efficient tools for describing the average behavior of processes, but can provide only an isolated observation and a brief appraisal of a workforce system; thus they lack the ability to model highly dynamic systems including people entering and exiting through recruitment and attrition over time. Since regression models usually consider just one output, they may be complicated or not suitable for modeling systems with several outputs.

3.6 | Analytical stock-and-flow models

Stock-and-flow models illustrate occupations or workforce cohorts, as stocks and movements between them (e.g., promotion, job rotation, and recruitments) as flows. One of the advantages of stock-and-flow modeling is that it can provide a systematic view of workforce structures and depict a workforce system in a manner that is easily interpreted by policymakers. It is also capable of incorporating with the concept of time by easily describing changes in the system over time.

Wilson (1987) described both supply and demand sides of a workforce planning model via stocks and flows. His model considers technology changes, competition for productivity increases relative to others in the field, governmental legislations, and social norms alterations. Sing et al. (2012a and 2012b) applied a stock-and-flow technique for a long-term forecast of labor supply in the construction industry, which considered the effect of economic growth on workforce supply for specific trade types. Fraher, Knapton, Sheldon, Meyer, and Ricketts (2013) developed a forecasting stock-and-flow model to predict the number of US surgeons by sex, age, and specialty from 2009 to 2028, to take into account policy factors as well as surgeons' demographics. Recently, Crettenden et al. (2014) found the stock-and-flow methodology best fitted their purpose in projecting health workforce of Australia to 2025. They analyzed the sensitivity of their model to parameters such as innovation, reform, and immigration, and examined how the outputs, such as future workforce shortage, would be different with changes in future. One limitation of stock-and-flow models is that they are highly dependent on computerized analytical solutions. However, this approach is the base of one of the most powerful simulation tools: system dynamics.

3.7 | Simulation models

Simulation is a methodology to test the behavior of a real system over time, usually via computers. Simulations define the functional behavior and attributes of the system by representing the logical or mathematical relations among the parts of that system. They may not have mathematical or algebraic solutions and are a suitable approach for investigating systems, which cannot be solved analytically (Law & Kelton, 1991; Winston, 1994). Simulation models are powerful approaches for answering "What if" questions, but can be expensive and time consuming to develop (Budnick, McLeavey, & Mojena, 1988), and their construction and the interpretation of their outputs may require the intervention of mathematical or statistical experts.

Depending on the type of the system being simulated and the purpose of the simulation, different dimensions of simulation can be used (Banks, Carson, Nelson, & Nicol, 2001; Law & Kelton, 1991):

- stochastic (probabilistic) or deterministic—depending on whether model input variables and parameters and hence the outputs are known with certainty or the inputs are probabilistic; hence the outputs can be considered only as estimates of the true characteristics of a model;
- continuous or discrete—depending on whether model states can change continuously over time or only at a discrete set of points in time; and
- steady state or dynamic—based on whether they consider system evolutions continuously over time or at a particular time point.

Based on the general framework for mapping a simulation model to its real-world system, there exist three major simulation methods:

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- discrete event simulation, which simulates a process as a series of events as time passes;
- agent-based simulation, which simulates the interaction among autonomous agents following predefined rules to achieve their objectives; and
- system dynamics (SD), which simulates systems as a series of stocks and flows rather than the individual behavior of agents.

Since very few agent-based and discrete event simulations have been developed for workforce prediction modeling, we focus on SD in this review. SD is a methodology for modeling the uncertain and nonlinear behavior of complex systems. It is a continuous system modeling approach which came from engineering control theory and uses stocks and flows, time delays, and internal feedback loops. The mathematical model of the overall SD structure is a system of first-order differential and integral equations (Sterman, 2000).

Previous use of this method includes a simulation model by Song and Rathwell (1993) to estimate the demand for doctors in China. They analyzed the effect of factors such as socioeconomics and demographic and health system determinants on demand for doctors based on target physician-to-population ratio. Park, Lee, Yoon, and Yeon (2008) proposed an integrated SD model to forecast the workforce requirements of the information security industry in Korea. They wanted to see how various policies could improve the effectiveness of their workforce. Similar to Song and Rathwell, one of the limitations of their model was that some of the policy-related variables of the model were based merely on ratio relationships that influenced the accuracy of the simulation results. Mouza (2010) showed the dynamic properties and evolution of a workforce system in an SD model and formulated an optimal control problem. She presented a solution method to approximate the optimal policy in a straightforward manner, without solving sequentially or backwards as is dictated by common methods. One limitation of her work is that she did not explicitly investigate the effect of the out-transition rates on system evolution. Barber and Lopez-Valcarcel (2010) produced a supply-and-demand/need SD model for medical specialists in Spain. They considered education, demographics, and labor market parameters in their model, and variables controllable by health planners such as the number of admission places for medical students and the mandatory retirement age were set as parameters to simulate different scenarios. The main limitation of their approach is that the need for providers is estimated based on assumed growth rates of the number of providers per capita. These growth rates are constant over time and do not take into account demographic changes in the population. They are also based on American estimates despite the model being developed for Spain. Fraher et al. (2013) produced a dynamic stock-and-flow model to project the supply of surgeons by sex, age, and specialty in the USA from 2009 to 2028. However, this study did not consider demand for surgery or the effect that supply itself would have on surgery. Masnick and McDonnell (2010) took an SD approach to link clinical staff skill mix planning to healthcare dynamics and addressed the interactions, feedbacks, and delays that affect the equilibrium of healthcare system considering need and supply. Although they developed a broad endogenous systems model of healthcare that takes into account real-world variables such as technology development and political and economic environments, they did not consider age categories of the workforce, nor did they include any suggestion for the optimal number of trainees to enroll. Vanderby et al. (2010) created a supply-and-demand-based SD model of the cardiac surgeon workforce in Canada and evaluated the oversupply or inadequacy of current workforce stemming from resident enrollment between 2008 and 2030. In this recent work the workforce age and productivity, along with delays in the system resulting from long training periods, were considered. Theirs was the first model to include feedback loops to provide insight into providers' productivity in reaction to the condition of the system and the impact on both the supply and demand of providers. However, this work does not model systems in which surgical waiting time and queueing problems exist.

4 | DISCUSSION

This review found the common techniques applied to workforce forecasting and analyzed the quality of the models considering model testing, sensitivity analysis and model implementation. All models include some simplifications, and intricacies of systems ought to be considered. Workforce modeling has grown over time, methods and models have become more complex, and research in this area has matured. Early models were typically qualitative and relied heavily on managerial experience or deterministic mathematical models. However, some modelers perceived the need for applying statistics in models to take into account stochastic factors and uncertainties in real-life systems; the deficiency associated with impractical applications of theoretical works in a qualitative sense led researchers to realize the dynamics of workforce systems and develop analytical stock-and-flow models. The models developed during past decades were simplified in a linear fashion. Linear differential equations, linear optimal control, and linear

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regression analysis were dominant approaches, and in stock-and-flow models the inflows and outflows had a proportional relationship with stocks. The characteristics of linear models changed over time to nonlinear systems, which were superior for representing real organizations.

It is also notable that much of the work did not incorporate feedback loops, which are an inherent feature of real workforce systems. For example, demand disturbances may instigate resource actions such as hiring or releasing workers. Likewise, studying delays associated with resource hiring and the oscillations resulting from simultaneous occurrence of delays and negative feedback or balancing loops have received little considerations. Omitting these effects impairs a model's ability to accurately reflect workforce dynamics.

In contrast to the supply-side focus of early research, in recent years the number of demand- and supply-based studies have grown to exceed other approaches. Yet fewer attempts have been directed to need- and supply-based models. Qualitative method is the dominating approach to develop need- and supply-based models, while statistics and regression are the popular techniques in demand-based models.

Transparency of models—that is, explaining a model in sufficient detail to enable a reader to reproduce it—is an essential factor for model evaluation. Well-written articles should describe assumptions, parameters, equations, and structure of models in a way that makes it possible to duplicate them. Some articles did not mention exactly what approaches they used. Some other articles did not clarify which type of—for example—simulation technique they used, nor included details of their approaches.

Another factor in model evaluations is sensitivity analysis. Almost a half of works have not conducted sensitivity analysis. Amongst the papers paying attention to sensitivity analysis, most articles have tested alternative scenarios by varying values of parameters; however, researchers could use evolutionary approaches to adapt the model to the effective value of dynamic parameters instead of treating the uncertain parameters as static. For example, by applying a distribution to uncontrolled dynamic parameters such as voluntary wastage rate, more accurate estimates can be achieved during model simulation. Moreover, sensitivity analysis counting the effect of technology and its progress may have on workforce performance has not been conducted. For example, in healthcare systems, medical device changes and innovations have a great effect on HRH productivity and the role of humans. Moreover, Construction and even Service Industries workforces are vulnerable to rapid changes in technology. A vast number of authors have merely mentioned sensitivity analysis in their articles, but have not clarified the strategies they have used for sensitivity analysis, and some others have not conducted sensitivity analysis on all variables.

Test of model validity is the source of another deficiency distinguished in the articles. Less than a half of articles have evaluated the validity of the developed models. Some models (e.g., Jaffry & Capon, 2005; Reeves & Reid, 1999; Sing, Love, & Tam, 2014; Williford & Chang, 1999) were just tested based on feedback from professionals and qualitative data collected from senior managers. In other cases, the results of models were compared to other techniques; for example, Mantel and Gaalman (1982), Özdemir-Akyıldırım and Talay-Değirmenci (2015), and some other authors have trusted their models based on agreement between their results and data extracted from other relevant publications (e.g., Denton, Gafni, & Spencer, 1994; Haase, Lewis, & Kao, 1996; Ho, 2010; Sing et al., 2012b; Stone & Fiorito, 1986). Some articles merely mentioned that the validity of the model has been conducted, but the test approach has not been specified (e.g., Grobler & Zock, 2010; Masnick, 2009). However, some models have been precisely tested based on actual patterns (e.g., Choudhury, Sarkar, & Mukherjee, 2002b; Dawson, Barrett, & Ross, 1990; Fagerstrom, Lonning, & Andersen, 2014; Lin et al., 2001), models' ability to replicate the past (e.g., Jorssen, 1989; Kinstler et al., 2008; Song & Rathwell, 1993), historic censuses over time (e.g., the work of Freed, Nahra, & Wheeler, 2003), fit tests, and residual error tests (e.g., Lin & Juan, 2010; Wang & Lin, 2007; Zhihong, Zhiguang, & Qi, 2003). Of those included in this review, 70 models used pilot tests and a second set of data for validation tests (e.g., Bee-Hua, 2000; Hurst, 2006; Murphy et al., 2013; Persad, O'Connor, & Varghese, 1995). Therefore, determining whether the models accurately represent reality or not should be considered by future workforce modelers, and model results have to be compared with observations obtained under the same conditions.

One of the persistent challenges operations researchers have faced is getting models implemented and reflecting change in the real world. Of the 275 articles reviewed, 45 clearly indicated the models presented had been used in real applications. Some articles are quite vague, with sentences like "this approach is applied to this Industry during a period of reduction in workforce," while many of the remaining articles do not mention whether the results of the developed models are used or are planned to be used in future. Comparing model adoption by industry, we found that 57% of articles in the Navy and Military area have been implemented in the real world and the results of the models have benefited policymakers. Among construction models, 23% of the models have been applied to solve real-world problems, while 19% of models in healthcare and 17% of workforce models in industry have

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been applied to real applications. This raises questions regarding factors affecting model implementation.

Key aspects such as clear description of methodologies, considering uncertainties, testing the validity of models, and the implementations of developed models are missing in some studies. In future research, the best approaches for testing the validity of models and dealing with uncertainties should also be considered in addition to choosing the best modeling methodology. This may help the future models be more reliable and more attractive for real applications.

5 | CONCLUSION

This review is undertaken to synthetize the literature on workforce forecast models. As shown, workforce forecast is applied to different industries and involves a vast range of operation research techniques that have been classified into eight categories: qualitative, time series, optimization, statistical, simulation, analytical stock and flow, generic mathematical models, and models using fuzzy logic, gray analysis, GA, and neural networks. These approaches are aimed at different features of workforce modeling. For example, qualitative techniques can help when past and present data are not available; time series and statistical techniques predict the future of systems considering that past and present trends continue, and optimization models are structured to find the best policies that can lead to particular goals. Simulation models are useful for complicated systems where there is no analytical solution. Specifically, SD technique is highly efficient for nonlinear dynamic systems, especially when feedbacks and delays are inevitable. Gray and fuzzy techniques are appropriate approaches faced with incomplete or uncertain information. Each approach is engaged with limitations, and a combination of different techniques can lead to well-developed models.

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ORCID

Anahita Safarishahrbijari http://orcid.org/0000-0002-2656-6796

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Anahita Safarishahrbijari is a Research Assistant in the Department of Computer Science at the University of Saskatchewan, Canada. Her research focuses on machine learning and modeling. She received her M. Sc. degree from K.N. Toosi University of Technology, Iran.

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